



Fig. 4: The RTSA's unique digital phosphor (DPX) display and frequency mask trigger (FMT) help quickly identify, capture and troubleshoot frequency-hopping signals

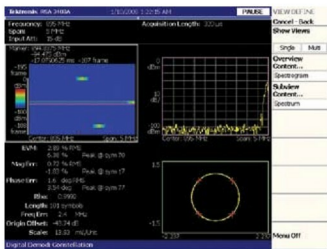


Fig. 5: Demodulating a captured off-the-centre hopped signal with a view of the spectrogram (upper left), frequency versus amplitude (upper right), signal modulation quality (lower left) and constellation (lower right)

main, a few RTSAs are able to produce a live RF view of the spectrum and provide a frequency mask trigger (FMT). These unique features simplify the troubleshooting of frequency-hopping signals.

A live RF view gives engineers a tool to instantly discover problems. In allowing users to view the actual signals for the first time, the latest RTSAs provide unmatched insight into RF signal behaviour. With spectrum updates that are at least 500 times faster than swept spectrum analysers, transient changes in frequency can be seen directly on the display. In the realm of SDR, this capability provides a com-

pletely new way to quickly assess the RF health of a signal and rapidly identify potential problems. Once a glitch or transient has been identified and defined as a frequency-domain event using a live, real-time view, the FMT can reliably capture the signal into memory for in-depth post-processing analysis. The frequency mask is user-defined and can be drawn to best capture the signal. With an infrequently occurring frequency hop, for example, the user is able to define the mask to trigger on the frequency excursion, rather than the change in power level. The frequency mask is defined as an envelope around this signal, and the instrument triggers once the signal enters the frequency mask area.

The combination of a live RF spectrum view and frequency triggering provides designers with a unique ability to find and troubleshoot problems frequently encountered with SDRs and the digital RF environment.

Modulation analysis of hopped signals

Modulation analysis of hopped signals across the full bandwidth requires an instrument that can not only trigger on and capture dynamic RF signals but also has the capacity for carrier-tracking vector analysis. Conventional vector signal analysers (VSAs) offer vector analysis for on-centre frequencies, but only very limited analysis of signals that are off-centre (i.e., 300 kHz or less). Most vector analysers lack the carrier tracking capability to demodulate the hopped signals across the full captured bandwidth.

Some RTSAs are capable of demodulating hopped signals across the entire capture bandwidth. Engineers are able to verify and debug their designs without having to assume the modulation quality at any off-centre frequency. One can choose to demodulate any of the captured signal hops, viewing time-correlated measurements from multiple domains with detailed modulation quality analysis.

Despite their ability to improve SDR performance, frequency-hopping techniques present unprecedented design and test challenges that conventional test instruments are unable to address. These radios require a new, flexible, integrated approach to SDR subsystem and system validation.

Leading-edge RTSAs deliver time-correlated measurements in multiple domains and the ability to see a live view of the RF spectrum. In addition, they provide an FMT, baseband IQ measurements and off-centre hopped signal demodulation. These capabilities simplify the testing and analysis of frequency-hopping radios, which are common in today's digital RF world. Working alone or in concert with other sophisticated test equipment, advanced RTSAs represent the most effective test solution for modern radio communication design, in-lab RF debug and in-field system evaluation. ●

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